

AlphaTichu

Toward a Superhuman Tichu Agent

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Abstract. We present AlphaTichu, the largest learned agent built to date for Tichu — a four-player, two-partnership, imperfect-information card game with a bidding layer and four special cards. The system is trained by behavioural cloning on **2.4 million** human games from the BrettSpielWelt server, validated by full action-level replay, conditioned on a per-player TrueSkill decile, and sharpened by gated self-play co-training of its play, card-passing, and bidding networks. This report covers two research campaigns. The first established a faithful expert imitator (play top-5 agreement **0.990** against top-decile humans) and found that of eight post-imitation methods — offline RL, PPO, determinized search, an AlphaZero-style loop, an asymmetric perfect-information critic, paired-advantage estimation, run-time adaptation, and anchored planning — exactly one produced a reliable gain (**+8.37 points/round**). The second campaign then closed every remaining self-play-gradient lever with pre-registered kills, and broke the resulting plateau from an unexpected direction: not a new algorithm, but a repaired *measurement* (the promotion gate's bootstrap had inflated its confidence intervals $\sim 1.45\times$, silently vetoing 129 consecutive true improvements) and a repaired *prior* (featurizer v7: a 233-dimension history block, the legal-action mask as a policy input, and a 39% forced-decision corpus prune). With those fixed, the *unchanged* co-training recipe banked six gate promotions and shipped a champion **+8.26 points/round** (95% CI [+7.18, +9.44], $n=81,920$) above the previously served agent. We distil the failures into three recurring causes — strategy fusion, self-inconsistent continuation targets, and an outcome-variance wall — and report, as a caution to practitioners, that our strongest agent briefly went live serving *random legal moves* behind a green health check. Whether AlphaTichu is superhuman remains, by construction, unverified: every instrument we have is self-relative, and we specify the human-in-the-loop protocol that would settle it.

1. Introduction

Games have long served as measured proving grounds for machine intelligence, and the frontier has moved from perfect-information board games — chess, Go, shogi [1,2] — to imperfect information: poker [3,4] and trick-taking card games [5,6]. **Tichu** sits at a particularly awkward point on that frontier. It is played by **four** players in **two fixed partnerships**; each player sees only their own hand; the deal is stochastic; and a high-variance **bidding** mechanic (Tichu and Grand-Tichu calls, worth ± 100 and ± 200 points) rides on top of the card play. Partners may not exchange information except through the legal moves of the game itself — cooperation must be learned, not signalled. The theory underwriting the strongest game AIs does not transfer: the CFR/ReBeL lineage is restricted in its guarantees to two-player zero-sum, and AlphaZero-style search assumes perfect information at the root. Tichu offers neither convenience.

This is the second edition of this report, and its shape is unusual. The first campaign (§7) built a faithful expert imitator and discovered, through eight methods and a great deal of compute, that only one of them beat it. The second campaign (§8–§10) set out to break the resulting plateau, killed every remaining reinforcement-learning lever under pre-registered criteria — and then broke the plateau anyway, through two things that are not algorithms at all: a repaired evaluation gate and a richer input representation. We believe the negative results, and the eventual diagnosis (*the plateau was a prior and a measurement problem, not an optimizer problem*), are the report's most durable contribution.

Contributions. (1) The largest imitation-learning Tichu system we are aware of: 2.4M games, replay-validated, skill-conditioned. (2) A compact 1,809-intent action space and a featurizer lineage that first compressed 16,568 dims to 224, then re-enriched to 824 on evidence. (3) A gated self-play co-training recipe that shipped two champions a full ship-bar apart. (4) A pre-registered, reproducible record of a dozen post-imitation methods, most of which failed, with a unifying diagnosis. (5) A candid account of the measurement and serving pitfalls — including a broken bootstrap that vetoed 129 true improvements and a champion that briefly served random moves behind a green health check.

2. The Game of Tichu

Tichu is played with a 56-card deck: ranks 2–A in four suits plus four special cards. The **Mahjong** (rank 1) leads the first trick and lets its player *wish* for a rank that others must play if able; the **Dog** passes the lead to one's partner; the **Phoenix** is a wildcard worth -25 ; the **Dragon** is the highest single, worth $+25$, but its winning trick must be given to an opponent. Each player receives 14 cards in batches of 8 and 6; after the first 8 a player may declare **Grand Tichu** (± 200), and before their first card, **Tichu** (± 100). Players then **Schupfen** — pass one card to each other player — and play tricks of singles, pairs, triples, full houses, stairs and straights, with **bombs** able to interrupt out of turn. If one team takes the first two finishes (a **Slam**) it scores $+200$ outright. Card points (Kings and Tens = 10, Fives = 5, Dragon $+25$, Phoenix -25) plus call bonuses accumulate until a team reaches 1,000.

Decision	Served by	Action space
Grand-Tichu Call	Call Network	binary (call / skip)
Schupfen	Schupfen Network	3 cards × 3 directions
Tichu Call	Call Network	binary (call / skip)
Play	trunk head	1,809 intents (incl. Pass)
Wish	trunk head	14 ranks
Dragon Assignment	trunk head	2 opponents

Table 1. The six decisions per round. Play, wish and dragon-assignment share one residual trunk; the two calls and Schupfen are standalone networks, all co-trained in the final system.

3. Data: the BrettSpielWelt Corpus

Supervision comes from **BrettSpielWelt (BSW)**, an online Tichu server with logs from 2007 onward: roughly **2.4 million games** and on the order of 1.5 billion decisions. Every game is subjected to **replay validation** — re-played action-by-action through an independent rules engine, whose computed score must match BSW's recorded result ($\geq 99.9\%$ reproduction) — turning a scraped dataset into a guaranteed-consistent one. A streaming **TrueSkill** sweep [12] rates every player handle; after a min-20-games cut, players are bucketed into **skill deciles 0-9** plus a neutral bucket, and the decile enters the model as a learned embedding, so one network spans novice to expert play. Conditioning on decile 9 is alone worth **+10.6 points/round** over the neutral bucket (95% CI [+3.95, +17.95]).

A finding of the second campaign reshaped the corpus itself: **39.4% of play decisions are forced** — exactly one legal intent — and contribute provably zero gradient under a masked softmax. Dropping them at materialisation time cut the play corpus from 1.248 billion to **756,291,715** rows and made the 824-dim v7 bundle (804 GiB) fit on disk at all. A tempting corollary — rescaling the play-head learning rate to compensate — was tested and *retracted*: Adam is scale-invariant to it (measured 3×10^{-8} parameter difference), and the 'compensation' would silently have slowed the play head by 39%.

4. Representation: Intents and the Featurizer Lineage

Tichu's concrete, suit-aware move set runs to $\sim 10,000$ entries, most strategically interchangeable. We train and predict over an **intent** action space of exactly **1,809** entries — one per (combination shape, rank, Phoenix position) tuple — and defer concrete card choice to a deterministic **Resolver**. The second campaign found and fixed a genuine defect here: intents whose realisations differ (one straight had 32 concrete variants, 30 of which shred a straight-flush bomb) were resolved by *set iteration order* — no decision at all. A lexicographic cost (bombs lost, suit-adjacency lost) now breaks ties in both the serving and rollout resolvers. It fires on 0.43% of decisions and is EV-neutral at our power (+0.73, CI [-4.27, +5.34]) — shipped as a correctness fix, expressly *not* claimed as strength.

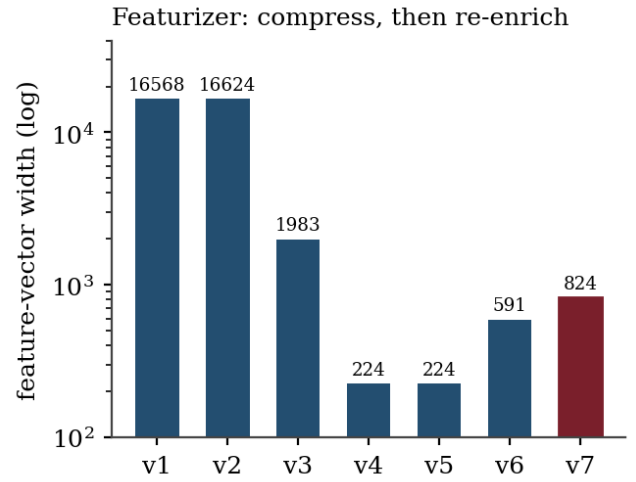


Figure 1. The featurizer lineage: aggressive compression (16,568 $\rightarrow$ 224 dims with no strength loss), then evidence-driven re-enrichment. v7 (824 dims) adds a 233-dim Rich History Block; its first 591 dims remain byte-identical to v6, which is what lets older champions serve as evaluation opponents through a prefix view.

The featurizer's history is itself a finding (Figure 1). v1-v4 compressed 16,568 dims to **224** with no measurable strength loss — capacity was never the bottleneck. v6 (591) added per-opponent played-by-planes. **v7 (824)** is the re-enrichment the belief post-mortem (§8) demanded: a **233-dim Rich History Block** — who was winning when an opponent declined, declined combo type / length / points / bomb flag, per-seat lead ranges, per-card play order, and a wish-void signal split into *proof* versus *evidence* — strictly additive, with the v6 prefix pinned byte-identical by golden fixtures. Alongside it, the 1,809-wide **legal-action mask became a policy input** (trunk input $824 + 64 + 1,809 = 2,697$), justified by a pre-registered probe: +3.04% relative NLL against a *negative* shuffled-mask placebo. One honesty note: the Rich History block's own residual pre-check was never completed — it shipped on an assumed positive, the weakest premise in the v7 design, redeemed only by the end-to-end result.

5. Architecture

A single **residual trunk** — four pre-norm residual blocks, width 1,024, GELU, projecting to 512 dims — feeds three heads: play (1,809 logits), wish (14), dragon_assignment (2). The feature vector is concatenated with a 64-dim skill embedding (11 rows) and, at v7, the legal mask. Standalone networks serve the two **Calls** and **Schupfen** (three masked 56-way heads, greedy-decoded). A **Belief Model** — a 3×56 occupancy grid over opponents' cards — exists off the serving path and was ultimately the instrument of its own programme's elimination (§8). Trunk capacity is closed *permanently*: a $2.25 \times$ sweep moved imitation accuracy +0.0008 in the first campaign, and a 2048×8 co-training control in the second sat exactly on the 1024×4 band at $4 \times$ the cost. The v6 $\rightarrow$ v7 migration widened the call networks function-preservingly (new columns zero-initialised; verified 0.0 logit difference on all four nets).

6. Training and the Imitation Ceiling

Behavioural cloning is masked cross-entropy over a memory-mapped, pre-featurised bundle (~130k examples/s; an epoch in hours on one H100). On held-out decile-9 games the master imitator reaches **top-1 0.834 / top-5 0.990** on play — the residual ~16% being the Bayes ceiling of imitation, where identical private states legitimately map to different expert moves. Behaviourally it matches decile-9 humans (Tichu-call rate 0.117 vs 0.145; bombs/round 0.115 vs 0.110). Offline refinement (AWR) was flat; capacity and data sweeps (Figure 2) show why: the constraint is the labels, not the model. Everything beyond this point requires signal that is not in the corpus.

The imitation ceiling is label-bound, not capacity- or data-bound

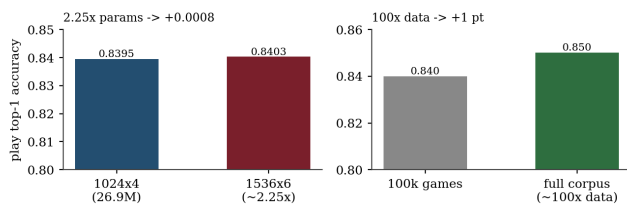


Figure 2. The imitation ceiling. 2.25× parameters buys +0.0008 play accuracy; ~100× data buys about one point. Label-bound, not capacity- or data-bound — the motivation for leaving the offline regime.

7. The First Campaign (to June 2026)

The first attempt to exceed the ceiling ran eight methods; Figure 3 (lower half) scores them. **PPO on the play head** was flat (+2.08, CI spanning zero). **Determinized search (PIMC)** got more passive with more search (1.8× the baseline's caller-passivity at deep budgets) — its leaf critic, regressed from a passive policy's self-play, prefers ceding, and faithful search amplifies that. An **AlphaZero-style search-and-learning loop** diverged monotonically (caller-passivity 0.182→0.478 over ten generations; EV collapse to −241). An asymmetric **perfect-information critic** came back flat, exposing the variance wall (§11). **Run-time adaptation (pMCPA)** lost even with oracle knowledge of the true deal (−67.83); **piKL anchored planning** provably maximised its own rollout objective and lost −58.66 at the table; a **re-anchored vine** collapsed to −23.59. Deterministic **claim-solver probes** showed the baseline already converts its guaranteed outs and is right to conserve bombs (force-bombing loses −17.4) — dismantling the behavioural story for an easy gap.

The one success, **Full-Stack Co-Training**, trains the play head, Schupfen, and both call networks together in self-play against a **frozen anchor**, with a perfect-information critic as variance reducer and per-net KL leases: **+8.37 points/round** over the served master (95% CI [+6.22, +10.45], n=40,000), roughly half from calls and half from card play. Successive refinements of that lineage (a wish-head fix, PBRs-named variants) produced the served export cpfix3328 (≈+8.8 cumulative), which became the baseline every second-campaign method had to beat — and, for a month, nothing did: the follow-on lineage spent 27,000 iterations at parity with it (+0.27, CI [−0.84, +1.32]).

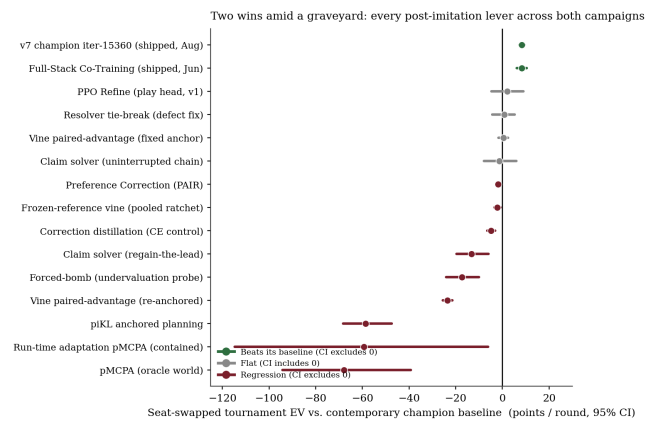


Figure 3. Both campaigns as one forest plot: every post-imitation lever, scored as seat-swapped tournament EV against its contemporary champion baseline (the June entries against the then-served master; the second-campaign entries against cpfix3328 or its co-train successor). Two methods clear zero — both are the same gated co-training recipe, a year of falsified alternatives apart. Note that several killed methods demonstrably optimised their own internal objectives while losing at the table.

8. The Second Campaign, Part I: Closing the Last Levers

July 2026 was spent killing, under pre-registered criteria, everything that remained. **Featurizer-by-intuition died first**: a proposed trick-stakes feature read +1.16% on a residual probe — and 0.4–0.6% once early stopping and a capacity sweep were applied, below the pre-registered 1% bar. The artefact taught a discipline (*no probe verdict without early stopping and a capacity sweep*) that every later greenlight had to survive. Half the proposal was also simply false: the engine's trick leader is the current winner — verified over 105,460 contested rows with zero counter-examples.

The frozen-reference vine — the principled successor to the re-anchored collapse, pairing branch playouts under a reference field advanced only on gate promotion, with pooled margins so small true gains could bank — was safety-stopped in CI-clean regression (pooled −2.19 [−3.48, −0.84], n=98,304; zero promotions). A three-round autopsy found no bug; a chimera head-swap convicted the trunk itself (trunk-only transplant: −2.53). With the fixed-anchor variant flat and this one negative, the paired-playout family's best observed gradient is zero, and “a pooled ratchet cannot bank zeros.” Family closed.

The belief programme died mechanically. A pre-registered EV gate asked what opponent-card modelling is worth in tournament points, by comparing a belief-guided chooser against a uniform card-counting one. The answer: the two arms are *the same agent 98.6% of the time* (they diverge on 1.41% of decisions; absurd upper bound ≈7 points/round, realistic well under one). The belief net beats the card-counting floor by +2.4 accuracy points with calibration indistinguishable from it, across every capacity — a ceiling, not a shortfall. The kill scopes to *card* modelling; its post-mortem — five history signals that consistently moved hand prediction — became the v7 Rich History Block. A literature deep-dive concurred: every privileged-information result we

could find is a *sample-efficiency* result, and this project is not sample-limited; trick-taking games have structurally low belief headroom (a cheating Skat sampler plays worse).

Preference Correction reopened the mined-blunder thread with its two known defects fixed: a pairwise sign-verified ordering (rank the verified better move above the chosen one; keep verified non-corrections ordered the other way) instead of blanket cross-entropy, trained where the damage was measured rather than where the KL guard sampled. The mechanism *confirmed* — same transfer at half the damage (5.6% vs 9.8%, $\sim 7\sigma$) — and the tournament killed it anyway: **-1.81** [-2.43, -1.21] at $n=81,920$, versus -4.81 for the cross-entropy control. A 0.17% behaviour change produced a CI-clean regression: the “small updates are exempt” escape from the self-inconsistency family (§11) is refuted, and the family gains its fourth member. The ADR also retracts two of its own motivating numbers (a “2.3× blunder-rate rise” was $1.56\times$ with the pool statistically unchanged, $p=0.211$; the stratified rate estimator reads $1.6\times$ high) — both errors, we note, had pointed in the direction of reopening.

Calls are not a lever. Chimera experiments transplanting call heads between the candidate, served, and BC agents all straddle zero: call heads are interchangeable, because the call decision sits at a flat EV optimum (marginal calls succeed at $P\approx 0.5$). The call-bonus deficit against the served champion is inherited from the BC prior (-10.58), and co-training had improved cards *and* calls together (+10.99 / +3.50) — no trade-off existed. The lever list was empty.

9. Part II: The Gate Was Broken

The call investigation surfaced something better than a lever: **the promotion gate itself was defective**. It had been bootstrapping *paired* seat-swap deltas as if unpaired, inflating its confidence intervals $\sim 1.45\times$ — so a promotion effectively required +4.8 points/round when the recipe's true step size is +0.5-2. The learner had won 129 consecutive evaluation windows without one promotion firing. What looked for a month like an inescapable local optimum *was the confidence interval*. The repaired gate — paired cluster bootstrap, verdicts pooled across windows, pooling bounded to the last 10 windows (unbounded pooling lets two bad windows impose ~ 3.7 points of permanent debt), and the served champion held out of the training rollout as a **required, out-of-distribution ship bar** — resolves ± 1.17 points at $n=40,960$ where the old one resolved ± 4.7 at 4,096. Small true gains became bankable, and every v7 number below is certified by this instrument.

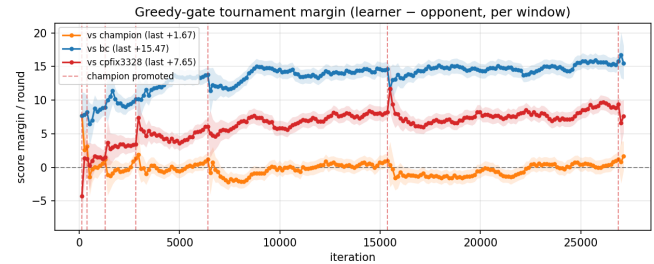


Figure 4. The repaired greedy gate at work over the full v7 run: per-window tournament margin of the learner against its three evaluation streams — the rolling champion (orange), the v7 BC prior (blue), and the served export cpfix3328 (red, held out of the training rollout as the required ship bar). Dashed lines mark gate promotions. The three streams reconcile (champion $\approx$ cpfix3328 + ~ 8 by either route), the ratchet banks small steps without regressions, and the run continues past the shipped iter-15360 champion.

10. Part III: v7 and the Second Champion

The v7 run changed the prior and the measurement — and *nothing else*. The co-training recipe (KL targets, gate cadence, PPO settings) is byte-identical to the run that had just spent 27k iterations at parity. New: the 824-dim features with the legal-mask input, a BC warm-start retrained on the pruned 756M-row corpus, function-preservingly widened call networks, and one repaired warm-start (the previous lineage had initialised Schupfen from a stale checkpoint that was simultaneously its anchor and its gate floor — a concrete candidate for that lineage's Schupfen deficit).

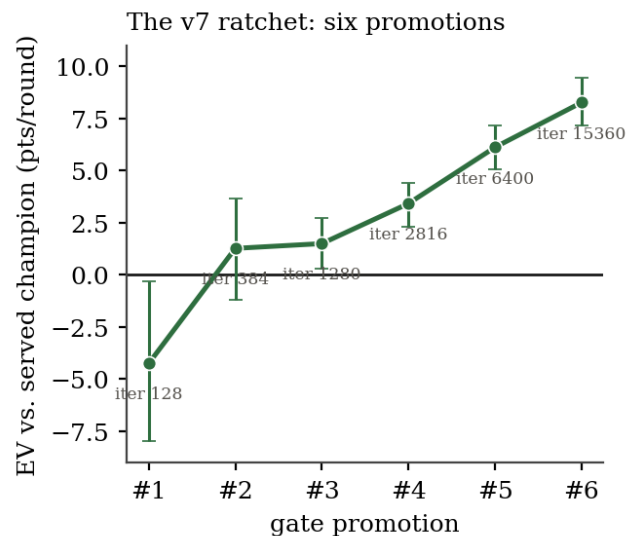


Figure 5. The v7 ratchet: EV of each gate promotion against the served champion cpfix3328 (95% CI, n up to 81,920). Promotion #1 beats its own lineage while still 4 points behind the served bar; promotion #3 is the first CI-clean clear of the served export by any lineage; promotion #6 ships at +8.26.

Six promotions followed (Figure 5). Promotion #3 (iter 1,280) was the first CI-clean clear of the served export by *any* lineage (+1.49, ci-lo +0.28, $n=57,344$); the shipped **promotion #6 (iter 15,360)** measures **+8.257 points/round over the served champion** (95% CI [+7.175, +9.444], $n=81,920$ seat-swapped observations) and +14.67 over the v7 BC prior. Three checks defend the number against promotion-selection bias: the served-champion stream showed no spike into the promoting window (8.02-8.26 across five readings); the three evaluation streams reconcile to

the same margin by independent routes; and the served champion is excluded from the training rollout, making the margin an out-of-distribution measurement the learner was never trained to inflate. Because the baselines differ, the two campaigns' headline numbers do not add arithmetically; what is claimed is one full ship-bar step ($\approx +8$) beyond the agent that the entire previous programme ended at parity with. The run has since posted a further uncommitted promotion (+9.36 at iter 26,880), unshipped pending gate certification.

Attribution within v7 is deliberately deferred — the isolation arms (v6-prefix slice $\times$ mask on/off against the same bundle) are runnable but were not exercised before shipping. What is certain is the shape of the explanation: the same optimizer that could not move for 27,000 iterations delivered +8.26 when the representation and the measurement changed. The plateau was never about the gradient.

11. A Unifying Account of Failure

Across both campaigns, three failure families absorb every flat or negative result in Figure 3 (Table 2). The constructive corollary is precisely the shipped recipe: a frozen anchor (no self-play co-drift), a single info-set policy rather than per-world search (no strategy fusion), a critic used only for variance reduction (respecting the wall) — and, the second campaign's addition, an evaluation instrument precise enough to bank the small true steps such a conservative recipe produces.

Strategy fusion	Determinized-world methods that take per-world argmaxes (PIMC, the search-and-learning loop) implicitly assume the hidden cards become known, over-valuing information-dependent plans. 96–99% of ‘true-world’ wins evaporate when marginalised across worlds.
Self-inconsistent targets	Methods whose target is computed under a continuation the deployed agent will not follow: piKL's anchor rollout, pMCPA's frozen advantage, the re-anchored vine, and — closing the ‘small update’ escape — Preference Correction, where a 0.17% behaviour change cost a CI-clean –1.81.
The variance wall	Outcome variance in stochastic self-play is dominated by the sampled future, not the hidden hands; ~55–60% is irreducible even for a perfect-information critic, and resolving a –2/round effect at per-state pairing still needs ~30k games. Privileged information buys sample efficiency — and this project is not sample-limited.

Table 2. The three recurring failure families, updated with second-campaign members.

12. Serving, and Two Cautionary Bugs

The agent is served by a stateless FastAPI service loading TorchScript artifacts, with version pins enforced at startup, four difficulty tiers (a rule-based easy; one checkpoint conditioned on ascending skill deciles for the rest), CPU-only inference against a p99 < 500 ms budget, and a guarantee that every response

is a legal action — on model failure the service falls back to a random legal move rather than stalling a game. That last guarantee produced the most instructive bug of the project: a flag lost in a shared-model code path made the v7 champion's forward pass raise on every call, so the fallback answered instead — **the strongest agent we had ever built went live serving uniformly random legal moves, with HTTP 200s, plausible-looking play, and a green health check**. It was caught only by exporting the real artifact and inspecting actual responses. A second bug in the same family: the decision-trace logger *re-ran* the forward pass it was meant to record, outside the fallback guard — converting a gracefully-degraded move into a 500 exactly when the trace mattered most. Both are now fixed (memoised forwards; explicit fallback records), and both argue the same lesson: in systems that guarantee liveness, *degradation is silent by design*, and only end-to-end inspection of served behaviour catches it. A residue remains: until the game client ships v7 fields, the champion serves with its 233 new dimensions zeroed — below its measured strength, and silently.

13. Is AlphaTichu Superhuman?

Unchanged in kind, and it matters that this is stated plainly: **every number in this report is self-relative** — v7 versus cpfix3328, cpfix3328 versus master, master versus its zoo. The tournament instrument can prove one agent beats another; it cannot prove an agent beats a human. The naive human experiment is statistically hopeless at these effect sizes (a human's 100–200 rounds give a ± 48 -point half-width against sub-10-point effects), and two structural worries sharpen it: seat-swap self-play may be blind to exactly the coordination gains that matter with *human* partners, and the shipped agent currently plays below its measured strength at the wire. Blunder-mining bounds the remaining self-play headroom as real but thin — ~0.33 robust blunders/round against the current champion, a ceiling of a few points — and the literature review points the next lever at partner coordination rather than opponent modelling. The honest path to the title's claim is unchanged: a powered, duplicate-deal human protocol. It has still not been built; the second campaign chose — correctly, on the evidence — to spend its effort repairing the instruments it had.

14. Future Work

The programme's open threads are concrete, and most already have machinery waiting. Nearest-term: **certify and ship promotion #7** — the v7 run has continued past the shipped checkpoint and posted +9.36 [+8.22, +10.45] at iter 26,880 against the served baseline; it ships when (and only when) the gate certifies it. Equally cheap: **close the serving gap** — the champion plays with its 233 Rich History dimensions zeroed until the game client transmits v7 fields, so the single largest immediate strength gain is a wire-format update, not a training run.

Attribution. The v7 gain is currently a bundle. The isolation arms are designed and runnable against the

same corpus — the v6-prefix slice crossed with mask on/off — and would apportion the +8.26 between the Rich History Block, the legal-mask input, and the repaired warm-start. The Rich History residual pre-check, abandoned mid-collection, should be completed retroactively: v7 shipped on an assumed positive, and the assumption deserves its verdict even after the fact.

Partner coordination. The literature review and our own coupling probes agree on where the remaining algorithmic headroom is *not* (opponent-card modelling, privileged critics, deeper search) and point instead at the one family this project has never tried: joint-policy-search methods for the *cooperative* half of the game, with bridge bidding [14] as precedent and Schupfen — the one decision whose coupling probes were decisively non-flat — as the natural target. This also demands an evaluation upgrade: seat-swapped self-play pairs each agent with a copy of itself, and is plausibly blind to exactly the convention-formation gains that matter with a human or heterogeneous partner; a cross-play harness should precede the method.

The remaining self-play headroom is bounded and mapped: ~0.33 robust blunders/round against the current champion, a ceiling of a few points, concentrated in policy-deterministic endgame states that only a playout can value. Two modest levers remain unexercised there: the claim solver as a *deployed* endgame module (so far used only as a probe), and gate-certified distillation of mined corrections under the pairwise mechanism — though \$8 counsels pessimism. The lower difficulty tiers also deserve honest treatment: medium and hard currently serve skill deciles the self-play programme never optimised.

The human instrument. Last and largest: the title's question still has no measuring device. The design is specified — duplicate-deal (mirror-match) play to cancel deal variance, expert human partnerships, enough rounds for sub-10-point resolution — and both closing ADRs of the second campaign named it as the successor once their levers died. Every alternative on this list is an optimisation; this one is the only item that can settle the claim the project is named for.

15. Conclusion

Two campaigns, a dozen methods, two shipped champions — both from the same conservative recipe — and a graveyard whose members increasingly died by pre-registered criteria rather than by surprise. The second edition's lesson is the one we least expected to write: after every algorithmic lever was closed, the plateau yielded to a fixed bootstrap and a richer input. We commend to practitioners the three failure families of \$11, the probe discipline of \$8, the gate design of \$9 — and the humility of \$12: measure the thing you actually serve. The superhuman question stays open, now for want of an instrument rather than an agent.

Availability

The shipped AlphaTichu agent can be played against, free of charge, at tichu.inzelabs.com. The complete

source code — engine, training pipeline, evaluation harness, and the ADRs and lab notes this report is drawn from — is available at github.com/ainze/tichu.

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